BODY CONTROL MODULE (BCM) INSPECTION

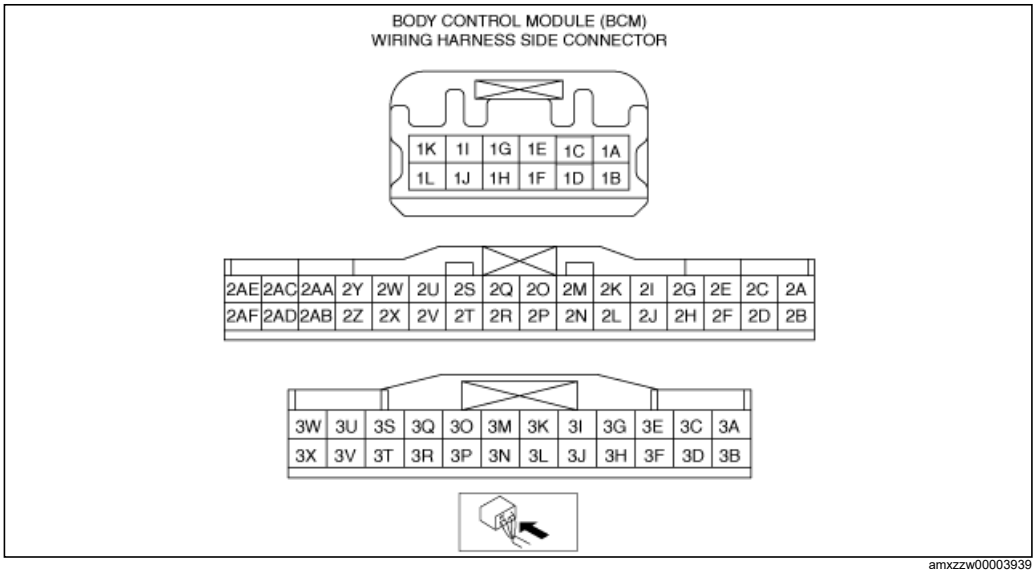
id094000003900

L.H.D

1. Disconnect the negative battery terminal. (See NEGATIVE BATTERY TERMINAL DISCONNECTION/CONNECTION.)
2. Remove the scuff plate (driver's side). (See SCUFF PLATE REMOVAL/INSTALLATION.)
3. Remove the front side trim (driver's side). (See FRONT SIDE TRIM REMOVAL/INSTALLATION.)
4. Remove the BCM. (See BODY CONTROL MODULE (BCM) REMOVAL/INSTALLATION.)
5. Connect the BCM connector.
6. Connect the negative battery terminal. (See NEGATIVE BATTERY TERMINAL DISCONNECTION/CONNECTION.)
7. Verify that the voltages of each of the terminals are as indicated in the terminal voltage table (reference). (See Terminal Voltage Table (Reference).)
   - If the voltage is not as specified in the terminal voltage table (reference), inspect the parts under Inspection item(s) and related wiring harnesses.
     — If the system does not work normally even though the inspection items are normal, identify the malfunctioning location according to the directions in [DIAGNOSTIC SYSTEM] and [SYMPTOM TROUBLESHOOTING] then perform repairs.

R.H.D

1. Disconnect the negative battery terminal. (See NEGATIVE BATTERY TERMINAL DISCONNECTION/CONNECTION.)
2. Remove the following parts:
   (1) Passenger-side scuff plate (See SCUFF PLATE REMOVAL/INSTALLATION.)
   (2) Passenger-side front side trim (See FRONT SIDE TRIM REMOVAL/INSTALLATION.)
   (3) Passenger-side A-pillar trim (See A-PILLAR TRIM REMOVAL/INSTALLATION.)
   (4) Passenger-side lower panel (See PASSENGER-SIDE LOWER PANEL REMOVAL/INSTALLATION.)
   (5) Selector lever knob (AT) (See AUTOMATIC TRANSMISSION SHIFT MECHANISM REMOVAL/INSTALLATION.)
   (6) Shift lever knob (MT)
   (7) Shift panel component (See SHIFT PANEL REMOVAL/INSTALLATION.)
   (8) Upper panel (See UPPER PANEL REMOVAL/INSTALLATION.)
   (9) Parking brake lever boot panel (See PARKING BRAKE LEVER BOOT PANEL REMOVAL/INSTALLATION.)
   (10) Rear console (See REAR CONSOLE REMOVAL/INSTALLATION.)
   (11) Front console panel (See FRONT CONSOLE PANEL REMOVAL/INSTALLATION.)
   (12) Front console component (See FRONT CONSOLE REMOVAL/INSTALLATION.)
   (13) Front cup holder bracket (See FRONT CUP HOLDER BRACKET REMOVAL/INSTALLATION.)
   (14) Foot support (See FRONT CUP HOLDER BRACKET REMOVAL/INSTALLATION.)
   (15) Woofer (See FRONT CUP HOLDER BRACKET REMOVAL/INSTALLATION.)
   (16) BCM (See BODY CONTROL MODULE (BCM) REMOVAL/INSTALLATION.)
3. Connect the BCM connector.
4. Connect the negative battery terminal. (See NEGATIVE BATTERY TERMINAL DISCONNECTION/CONNECTION.)
5. Verify that the voltages of each of the terminals are as indicated in the terminal voltage table (reference). (See Terminal Voltage Table (Reference).)
   - If the voltage is not as specified in the terminal voltage table (reference), inspect the parts under Inspection item(s) and related wiring harnesses.
     — If the system does not work normally even though the inspection items are normal, identify the malfunctioning location according to the directions in [DIAGNOSTIC SYSTEM] and [SYMPTOM TROUBLESHOOTING] then perform repairs.



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Terminal Voltage Table (Reference)

Terminal	Signal	Connected to	Measurement condition		Voltage (V)	Inspection item(s)
1A	Power supply	HAZARD 15 A fuse	Under any condition		B+	HAZARD 15 A fuse
1B	Power supply	• ROOM 25 A fuse • INTERIOR 15 A fuse	Under any condition		B+	• ROOM 25 A fuse • INTERIOR 15 A fuse
1C	—	—	—		—	—
1D*6	Brake light control	Brake light relay	Body control module (BCM) simulation item STOP_LMP_CTL is on		1.0 or less	Brake light relay
			Body control module (BCM) simulation item STOP_LMP_CTL is off		B+	
1E	Rear turn light (RH) control	Rear turn light (RH)	Ignition is switched ON (engine off or on)	Turn switch is in right turn position	1.0 or less ↔ B+	Rear turn light (RH)
				Turn switch is in position other than above	1.0 or less	
1F	—	—	—		—	—
1G	Rear turn light (LH) control	Rear turn light (LH)	Ignition is switched ON (engine off or on)	Turn switch is in left turn position	1.0 or less ↔ B+	Rear turn light (LH)
				Turn switch is in position other than above	1.0 or less	
1H	Interior light power supply	Map light	Under any condition		B+	Map light
1I	—	—	—		—	—
1J	—	—	—		—	—
1K	Ground	Body ground	Under any condition		1.0 or less	Body ground
1L	Ground	Body ground	Under any condition		1.0 or less	Body ground
2A	Lock input (door lock-link switch (driver's side))	Door latch and lock actuator (driver's side)	Driver's door is locked		1.0 or less	Door latch and lock actuator (driver's side)
			Driver's door is unlocked		Approx. 10	
2B	Unlock input (door lock-link switch (driver's side))	Door latch and lock actuator (driver's side)	Driver's door is locked		Approx. 9.5	Door latch and lock actuator (driver's side)
			Driver's door is unlocked		1.0 or less	
2C	CAN_L	CAN system related module	Terminal used for communication therefore determination based on terminal voltage is not possible.			
2D	CAN_H	CAN system related module	Terminal used for communication therefore determination based on terminal voltage is not possible.			
2E	Top lock switch signal	Top lock switch	Convertible top is open		1.0 or less	Top lock switch
			Convertible top is closed		Approx. 9.5	
2F	—	—	—		—	—
2G	Door latch switch (passenger's side) signal	Door latch and lock actuator (passenger's side)	Door (passenger's side) is open		1.0 or less	Door latch and lock actuator (passenger's side)
			Door (passenger's side) is closed		Approx. 9.0	
2H	Door latch switch (driver's side) signal	Door latch and lock actuator (driver's side)	Door (driver's side) is open		1.0 or less	Door latch and lock actuator (driver's side)
			Door (driver's side) is closed		Approx. 9.5	
2I	—	—	—		—	—
2J	Serial communication	Start stop unit	Terminal used for communication therefore determination based on terminal voltage is not possible.			
2K*1	Door lock switch signal	Door lock switch (RH)	Door lock switch is turned to unlock side (driver's side or passenger's side)		1.0 or less	Door lock switch (RH)
			Door lock switch is turned to lock side (driver's side or passenger's side)		Approx. 1.25	
			Other		Approx 5.0	
2L	Door key cylinder switch signal	Door latch and lock actuator (driver's side)	Door key cylinder is turned to unlock side		1.0 or less	Door latch and lock actuator (driver's side)
			Door key cylinder is turned to lock side		Approx. 1.25	
			Other		Approx 5.0	
2M	Lock input (door lock-link switch (passenger's side))	Door latch and lock actuator (passenger's side)	Passenger's door is unlocked		1.0 or less	Door latch and lock actuator (passenger's side)
			Passenger's door is locked		B+	
2N	—	—	—		—	—
2O	—	—	—		—	—
2P	—	—	—		—	—
2Q	Fuel sensor signal	Fuel gauge sender unit	Inspect fuel gauge sender unit and related wiring harness because determination based on terminal voltage is not possible. (See FUEL GAUGE SENDER UNIT INSPECTION.)			
2R	Fuel sensor ground	Fuel gauge sender unit	Under any condition		1.0 or less	Fuel gauge sender unit
2S	Trunk lid unlatch switch signal (exterior)	Trunk lid opener switch (exterior)	Trunk is opened		1.0 or less	Trunk lid opener switch (exterior)
			Trunk is closed		Approx. 4.5	
2T	—	—	—		—	—
2U	—	—	—		—	—
2V	—	—	—		—	—
2W	LIN communication	Theft-deterrent siren	Terminal used for communication therefore determination based on terminal voltage is not possible.			
2X	—	—	—		—	—
2Y	LIN communication	Electrical supply unit (ESU)	Terminal used for communication therefore determination based on terminal voltage is not possible.			
2Z*6	Brake switch signal	• Start stop unit • Brake switch (No.1 signal)	Brake pedal is depressed		B+	• Start stop unit • Brake switch (No.1 signal)
			Brake pedal is not depressed		1.0 or less	
2AA	—	—	—		—	—
2AB	—	—	—		—	—
2AC	Trunk lid latch switch signal	Trunk lid latch and release actuator	Trunk is opened		1.0 or less	Trunk lid latch and release actuator
			Trunk is closed		Approx. 9.5	
2AD	—	—	—		—	—
2AE	—	—	—		—	—
2AF	—	—	—		—	—
3A	Front turn light (RH) control	Front turn light (RH)	Ignition is switched ON (engine off or on)	Turn switch is in right turn position	1.0 or less ↔ B+	Front turn light (RH)
				Turn switch is in position other than above	1.0 or less	
3B	Front turn light (LH) control	Front turn light (LH)	Ignition is switched ON (engine off or on)	Turn switch is in left turn position	1.0 or less ↔ B+	Front turn light (LH)
				Turn switch is in position other than above	1.0 or less	
3C	CAN_H	CAN system related module	Terminal used for communication therefore determination based on terminal voltage is not possible.			
3D	CAN_L	CAN system related module	Terminal used for communication therefore determination based on terminal voltage is not possible.			
3E	Blower motor relay control	• Blower relay • HEATER 40 A fuse	Ignition is switched ON (engine off or on)	Blower motor is operated using fan switch	1.0 or less	• Blower relay • HEATER 40 A fuse

Terminal	Signal	Connected to	Measurement condition		Voltage (V)	Inspection item(s)
				Blower motor is turned off by pressing climate control unit OFF button	B+	
3F	Windshield wiper INT relay control	Wiper INT relay	Ignition is switched ON (engine off or on)	Windshield wiper switch is in position other than OFF	1.0 or less	Wiper INT relay
				Windshield wiper switch is in OFF position	B+	
3G	Windshield wiper motor (HI) control	Wiper HI relay	Ignition is switched ON (engine off or on)	Windshield wiper switch is in HIGH position	1.0 or less	Wiper HI relay
				Windshield wiper switch is in OFF position	B+	
3H	Horn relay control	Horn relay	Ignition is switched ON (engine off or on)	Horn switch is on	1.0 or less	Horn relay
				Horn switch is off	B+	
3I	—	—	—		—	—
3J	—	—	—		—	—
3J	Head light cleaner relay control	• Head light cleaner relay • H/CLEAN 20 A fuse	Ignition is switched ON (engine off or on)	Connect the M-MDS and turn on the "H/L_CLN_RY" using the body control module (BCM) simulation function	1.0 or less	• Head light cleaner relay • H/CLEAN 20 A fuse
				Connect the M-MDS and turn off the "H/L_CLN_RY" using the body control module (BCM) simulation function	B+	
3K	Rear window defogger relay control	Rear window defogger relay	Ignition is switched ON (engine off or on)	Rear window defogger switch is on	1.0 or less	Rear window defogger relay
				Rear window defogger switch is off	B+	
3L	—	—	—		—	—
3M	Autostop switch signal	Wiper motor	Ignition is switched ON (engine off or on)	Windshield wipers are stopped in position other than park	1.0 or less	Wiper motor
				Windshield wipers are stopped in park position	B+	
3N	Power supply	• ENG+B 7.5 A fuse • ROOM 25 A fuse	Under any condition		B+	• ENG+B 7.5 A fuse • ROOM 25 A fuse
3O	Ignition power supply	METER1 10 A fuse	Ignition is switched ON (engine off or on)		B+	METER1 10 A fuse
			The ignition is switched off (LOCK)		1.0 or less	
3P	—	—	—		—	—
3Q*2	Washer fluid level sensor signal	Washer fluid level sensor	Ignition is switched ON (engine off or on)	Washer fluid level sensor is less than MIN (washer fluid level sensor on)	1.0 or less	Washer fluid level sensor
				Washer fluid level sensor is MIN or more (Washer fluid level sensor off)	B+	
3R	Brake fluid level sensor signal	Brake fluid level sensor	Ignition is switched ON (engine off or on)	Brake fluid level sensor is less than MIN (brake fluid level sensor on)	1.0 or less	Brake fluid level sensor
				Brake fluid level sensor is MIN or more (brake fluid level sensor off)	B+	
3S	LIN communication	Current sensor	Terminal used for communication therefore determination based on terminal voltage is not possible.			
3T	Horn switch control	Horn switch	Ignition is switched ON (engine off or on)	Horn switch is on (long press)	1.0 or less	Horn switch
				Horn switch is off	B+	
3U	Trunk unlatch switch signal (interior)	Trunk lid opener switch (interior)	Trunk lid opener switch (interior) is on (trunk lid opener switch (interior) is pressed)		1.0 or less	Trunk lid opener switch (interior)
			Trunk lid opener switch (interior) is off (trunk lid opener switch (interior) is not pressed)		B+	
3V	—	—	—		—	—
3W*4	Bonnet latch switch signal	Bonnet latch switch	Bonnet is open		Approx. 9.0	Bonnet latch switch
			Bonnet is closed		1.0 or less	
3X	LIN communication	• Auto-light sensor/rain sensor*3 • Theft-deterrenet sensor*5	Terminal used for communication therefore determination based on terminal voltage is not possible.			

*1 :With door lock switch
*2 :With washer fluid level sensor
*3 :With auto light system/auto wiper system
*4 :With bonnet latch switch
*5 :With theft-deterrent siren
*6 :With smart city brake support (SCBS)